Activity 1.7 – Universal Shift Register and 8-bit LFSR

1. Design Approach:

Designed an 8-bit universal shift register and an 8-bit Linear Feedback Shift Register (LFSR) using Verilog HDL. The universal shift register supports hold, left shift, right shift and parallel load operations. The LFSR generates a sequence using feedback logic.

2. Verification Steps:

- Created a separate testbench.
- Tested reset operation.
- Verified parallel loading.
- Tested left and right shifting.
- Checked the hold operation.
- Verified LFSR state advancement.
- Simulated using ModelSim.

3. Results:

The testbench checks the shift register outputs and LFSR state changes. PASS/FAIL messages are displayed.

4. Issues Faced:

Correct selection of shift modes and feedback logic is important. The LFSR must not start from an all-zero state.

5. Conclusion:

Successfully designed and verified a universal shift register and an 8-bit LFSR.